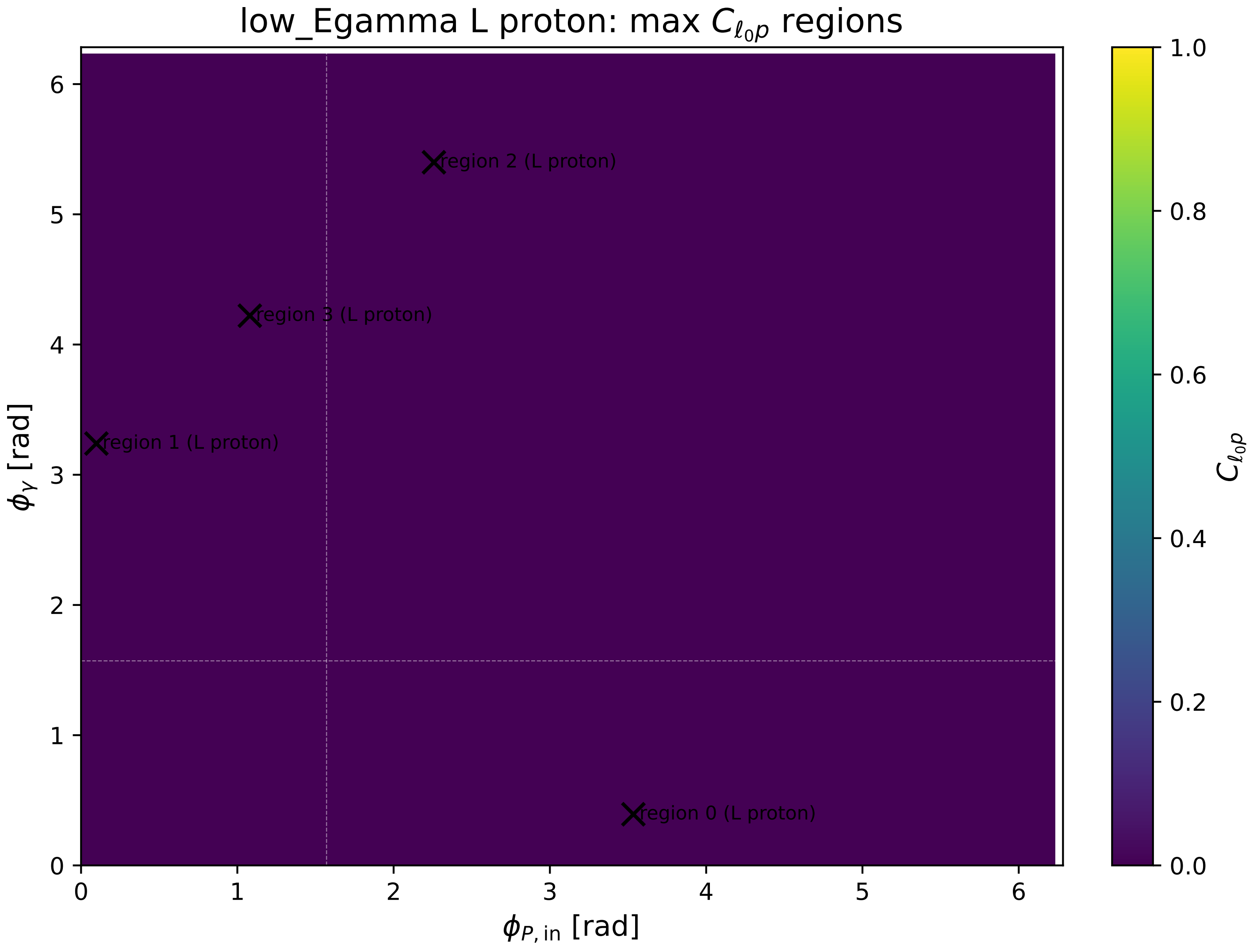
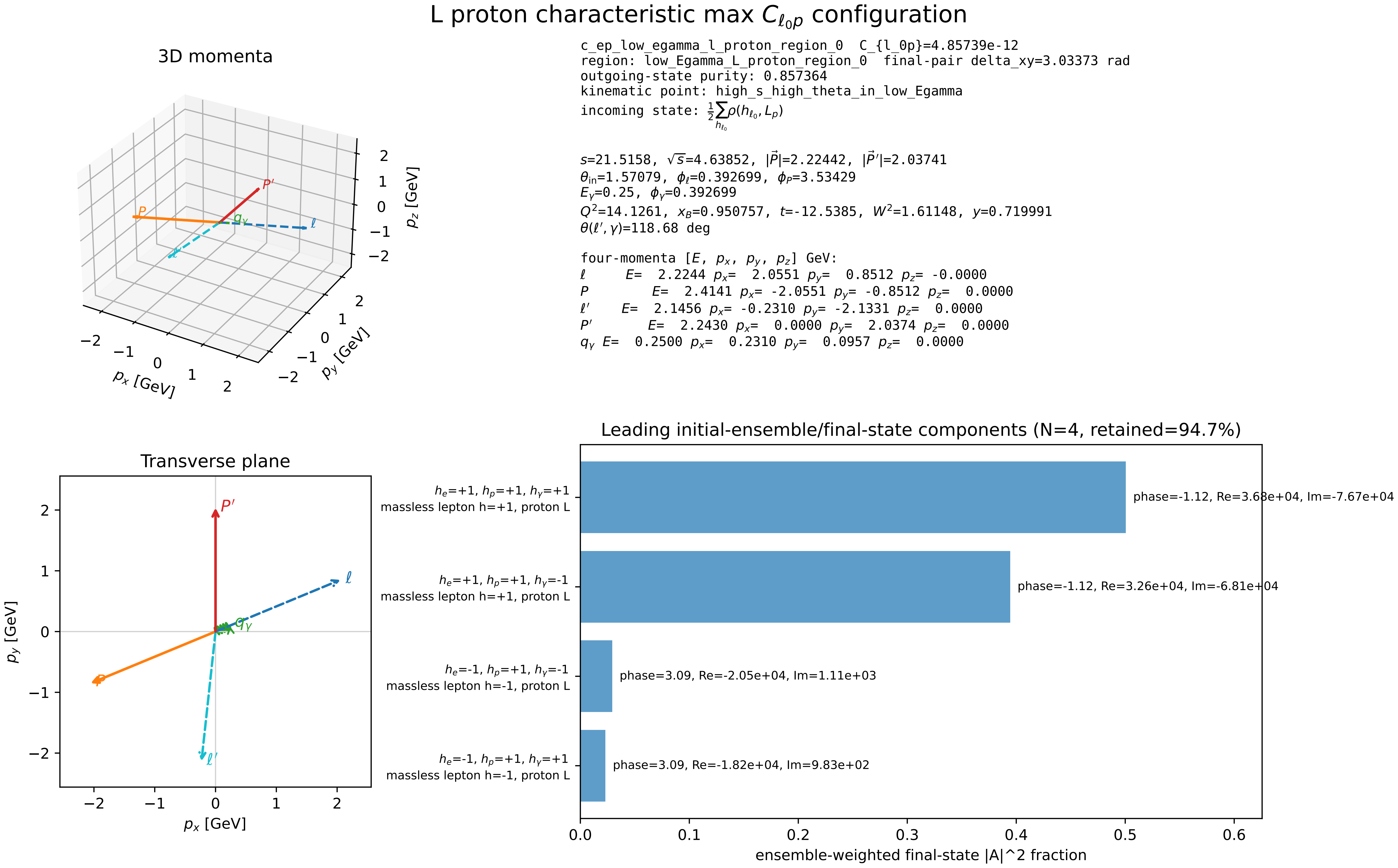
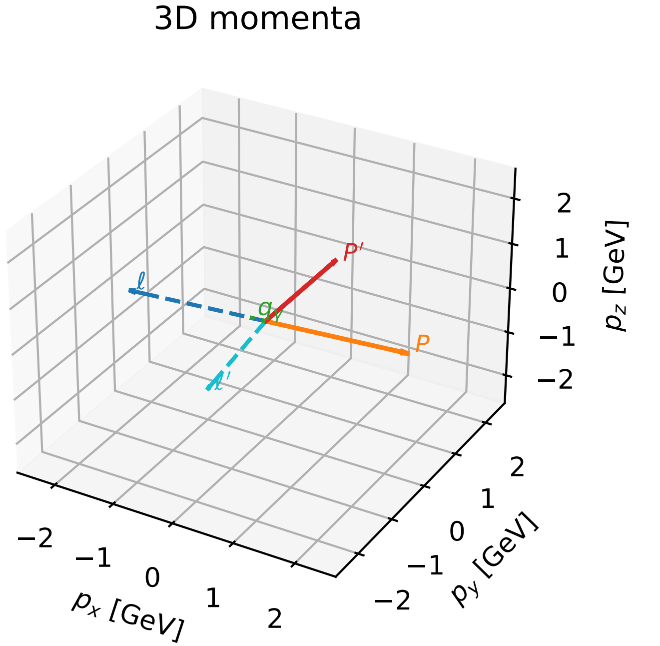


low_Egamma L proton: max $C_{\ell_0 p}$ regions
region 2 (L proton)
region 3 (L proton)
region 1 (L proton)
region 0 (L proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{\ell_0 p}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0



L proton characteristic max $C_{\ell_0 p}$ configuration
3D momenta
c_ep_low_egamma_l_proton_region_0 C_{l_0p}=4.85739e-12
region: low_Egamma_L_proton_region_0 final-pair delta_xy=3.03373 rad
outgoing-state purity: 0.857364
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, L_p)$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.03741
θ_{in}=1.57079, ϕ_ℓ=0.392699, ϕ_P=3.53429
E_γ=0.25, ϕ_γ=0.392699
Q^2=14.1261, x_B=0.950757, t=-12.5385, W^2=1.61148, y=0.719991
$\theta(\ell', \gamma)$=118.68 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 2.0551 p_y= 0.8512 p_z= -0.0000
P E= 2.4141 p_x= -2.0551 p_y= -0.8512 p_z= 0.0000
ℓ' E= 2.1456 p_x= -0.2310 p_y= -2.1331 p_z= 0.0000
P' E= 2.2430 p_x= 0.0000 p_y= 2.0374 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.2310 p_y= 0.0957 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P
P'
ℓ
ℓ'
q_γ
Transverse plane
Leading initial-ensemble/final-state components (N=4, retained=94.7%)
h_e=+1, h_p=+1, h_γ=+1
massless lepton h=+1, proton L
phase=-1.12, Re=3.68e+04, Im=-7.67e+04
h_e=+1, h_p=+1, h_γ=-1
massless lepton h=+1, proton L
phase=-1.12, Re=3.26e+04, Im=-6.81e+04
h_e=-1, h_p=+1, h_γ=-1
massless lepton h=-1, proton L
phase=3.09, Re=-2.05e+04, Im=1.11e+03
h_e=-1, h_p=+1, h_γ=+1
massless lepton h=-1, proton L
phase=3.09, Re=-1.82e+04, Im=9.83e+02
ensemble-weighted final-state |A|^2 fraction

L proton characteristic max $C_{\ell_0 p}$ configuration

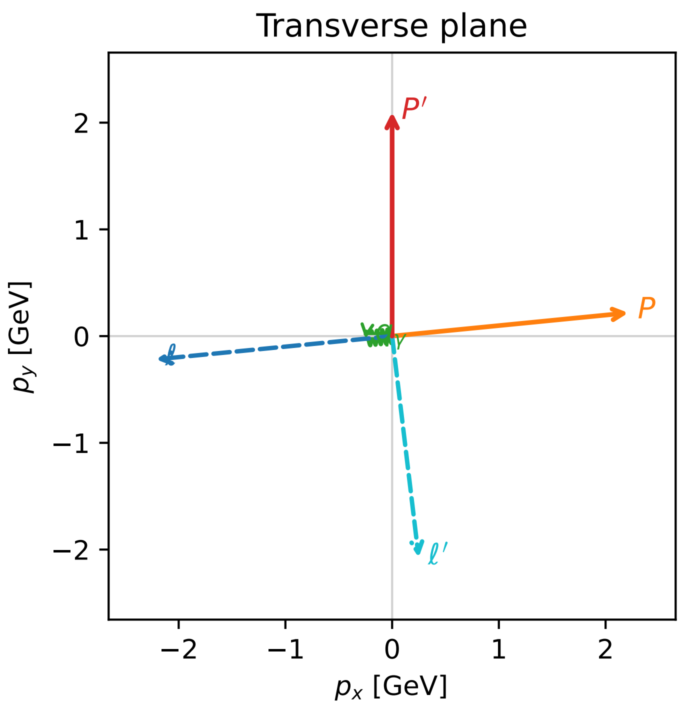


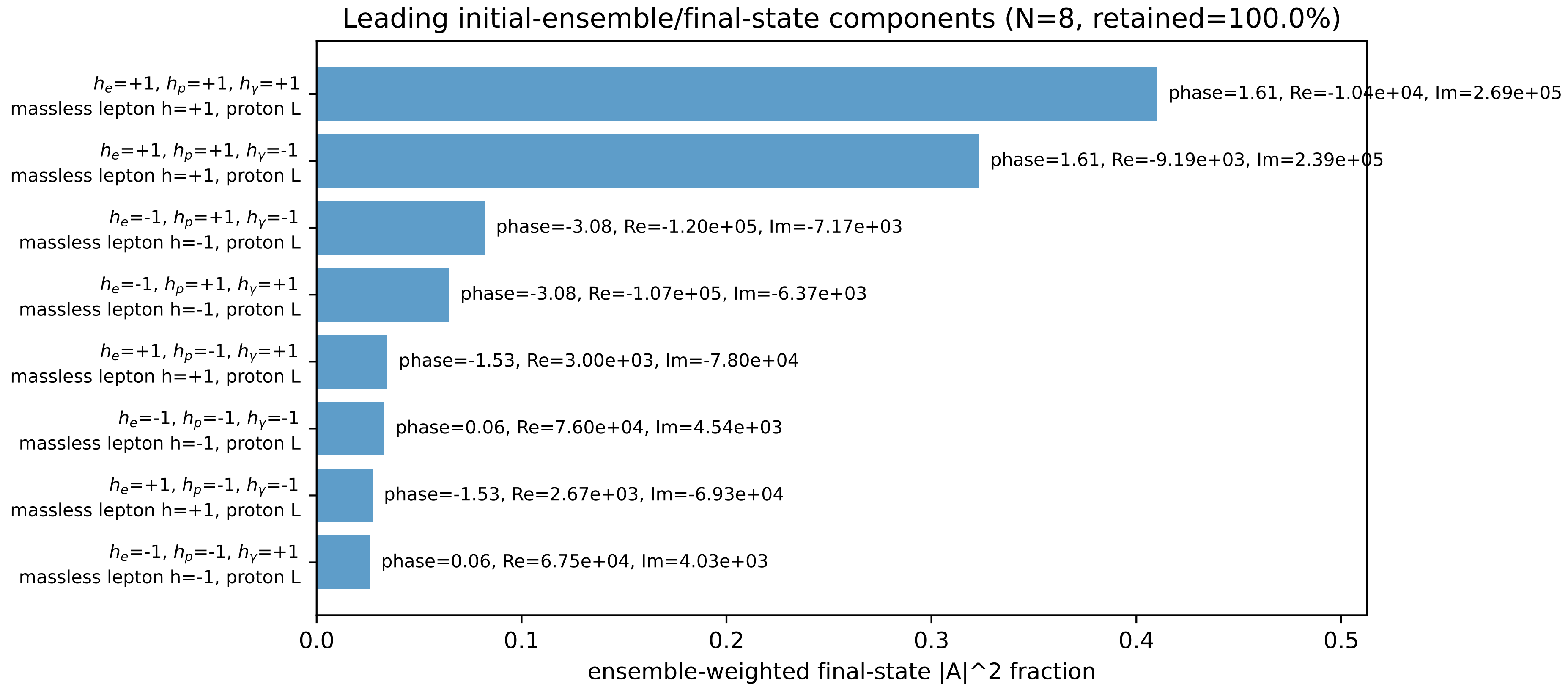

```
c_ep_low_egamma_l_proton_region_1  C_{l_0p}=3.60792e-12
region: low_Egamma_L_proton_region_1  final-pair delta_xy=3.02223 rad
outgoing-state purity: 0.673847
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_{\ell_0}} \rho(h_{\ell_0}, L_p)$

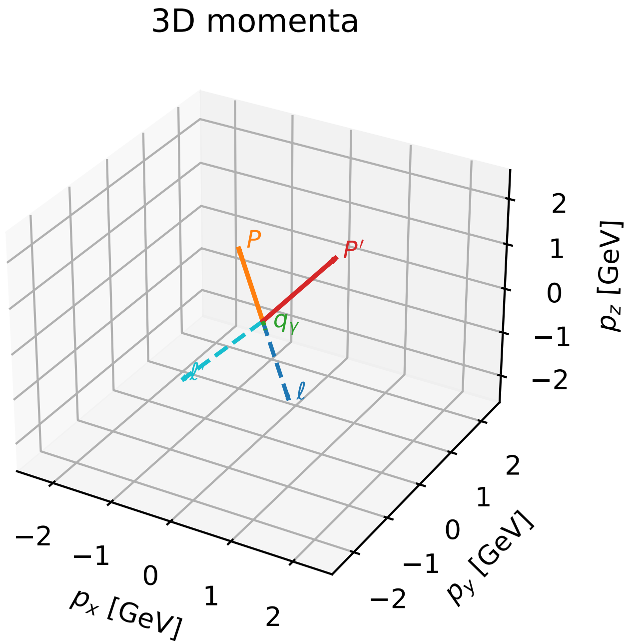
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.09905
$\theta_{\rm in}$=1.57079, ϕ_ℓ=3.23977, ϕ_P=0.0981748
E_γ=0.25, ϕ_γ=3.23977
Q^2=9.49235, x_B=0.883439, t=-8.42552, W^2=2.13227, y=0.520681
$\theta(\ell', \gamma)$=91.2137 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -2.2137 p_y= -0.2180 p_z= -0.0000
P E= 2.4141 p_x= 2.2137 p_y= 0.2180 p_z= 0.0000
ℓ' E= 2.0894 p_x= 0.2488 p_y= -2.0745 p_z= 0.0000
P' E= 2.2991 p_x= 0.0000 p_y= 2.0991 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2488 p_y= -0.0245 p_z= 0.0000





L proton characteristic max $C_{\ell_0 p}$ configuration

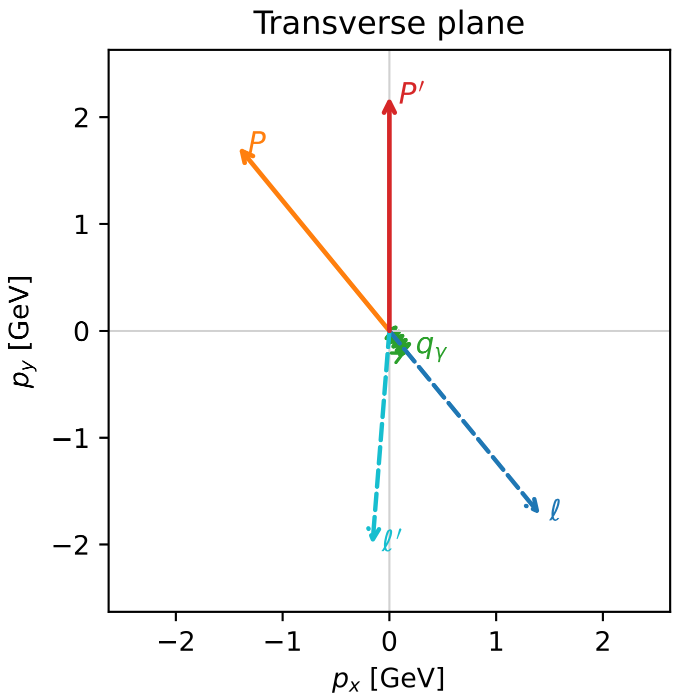


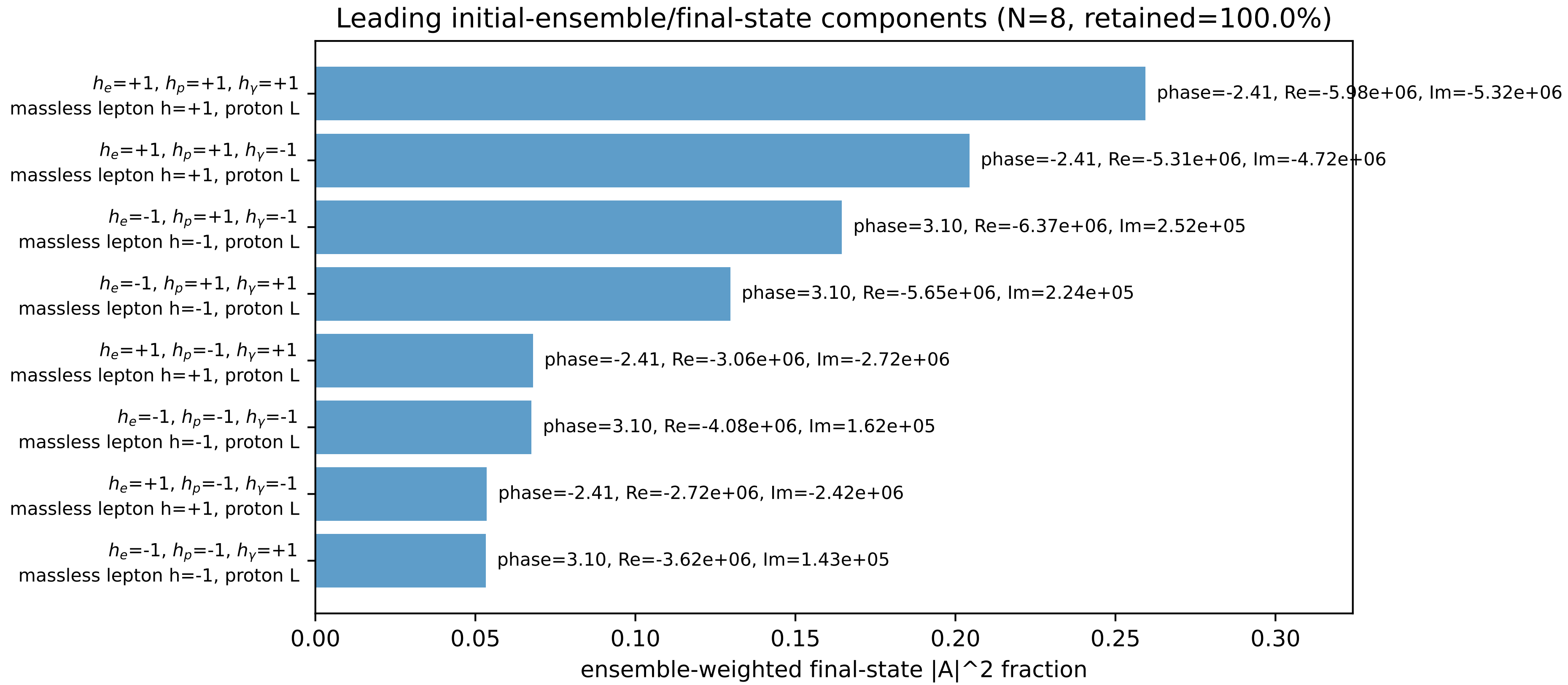

```
c_ep_low_egamma_l_proton_region_2  C_{l_0p}=3.09656e-12
region: low_Egamma_L_proton_region_2  final-pair delta_xy=3.06239 rad
outgoing-state purity: 0.514504
kinematic point: high_s_high_theta_in_low_Egamma
```

incoming state: $\frac{1}{2}\sum_{h_{\ell_0}} \rho(h_{\ell_0}, L_p)$

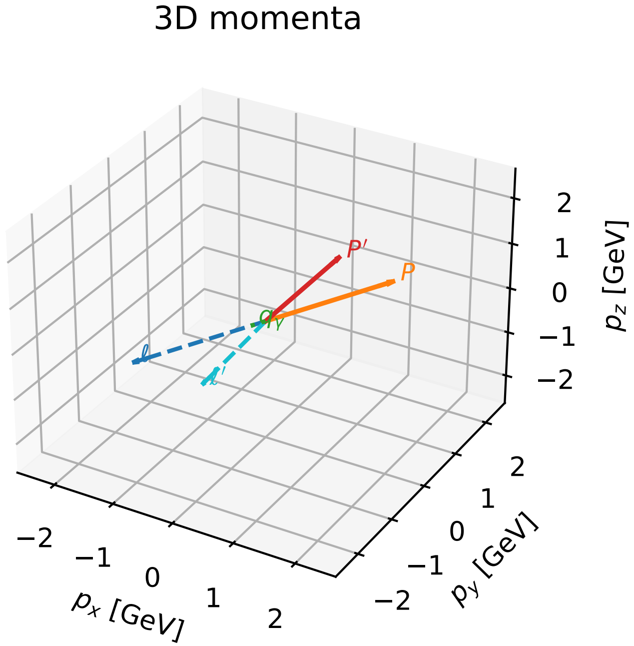
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.1916
θ_{in}=1.57079, ϕ_ℓ=5.39961, ϕ_P=2.25802
E_γ=0.25, ϕ_γ=5.39961
Q^2=2.49357, x_B=0.550144, t=-2.21332, W^2=2.91885, y=0.219644
$\theta(\ell', \gamma)$=43.9128 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= 1.4112 p_y= -1.7195 p_z= -0.0000
P E= 2.4141 p_x= -1.4112 p_y= 1.7195 p_z= 0.0000
ℓ' E= 2.0046 p_x= -0.1586 p_y= -1.9983 p_z= 0.0000
P' E= 2.3839 p_x= 0.0000 p_y= 2.1916 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.1586 p_y= -0.1933 p_z= 0.0000





L proton characteristic max $C_{\ell_0 p}$ configuration



c_ep_low_egamma_l_proton_region_3 C_{l_0p}=1.23066e-12
region: low_Egamma_L_proton_region_3 final-pair delta_xy=3.08234 rad
outgoing-state purity: 0.505113
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $\frac{1}{2}\sum_{h_{\ell_0}}\rho(h_{\ell_0}, L_p)$

s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22442, $|\vec{P}'|$=2.20723
$\theta_{\rm in}$=1.57079, ϕ_ℓ=4.22152, ϕ_P=1.07992
E_γ=0.25, ϕ_γ=4.22152
Q^2=1.30635, x_B=0.375519, t=-1.15953, W^2=3.05228, y=0.168579
$\theta(\ell', \gamma)$=31.5197 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2244 p_x= -1.0486 p_y= -1.9618 p_z= -0.0000
P E= 2.4141 p_x= 1.0486 p_y= 1.9618 p_z= 0.0000
ℓ' E= 1.9902 p_x= 0.1178 p_y= -1.9868 p_z= 0.0000
P' E= 2.3983 p_x= 0.0000 p_y= 2.2072 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.1178 p_y= -0.2205 p_z= 0.0000